

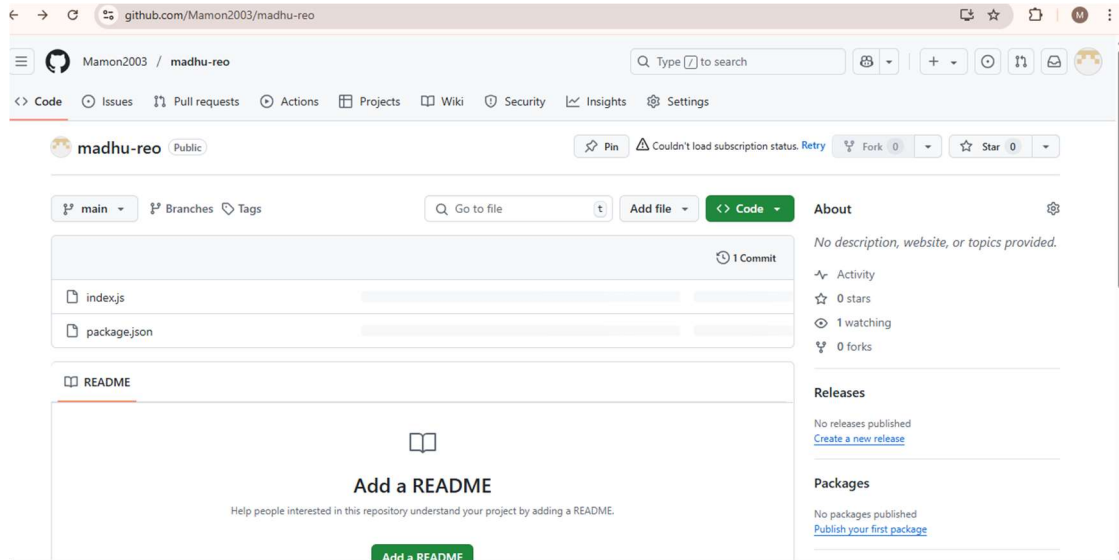
Assignment No: 09

Deploy a project from GitHub to EC2.

Solution:

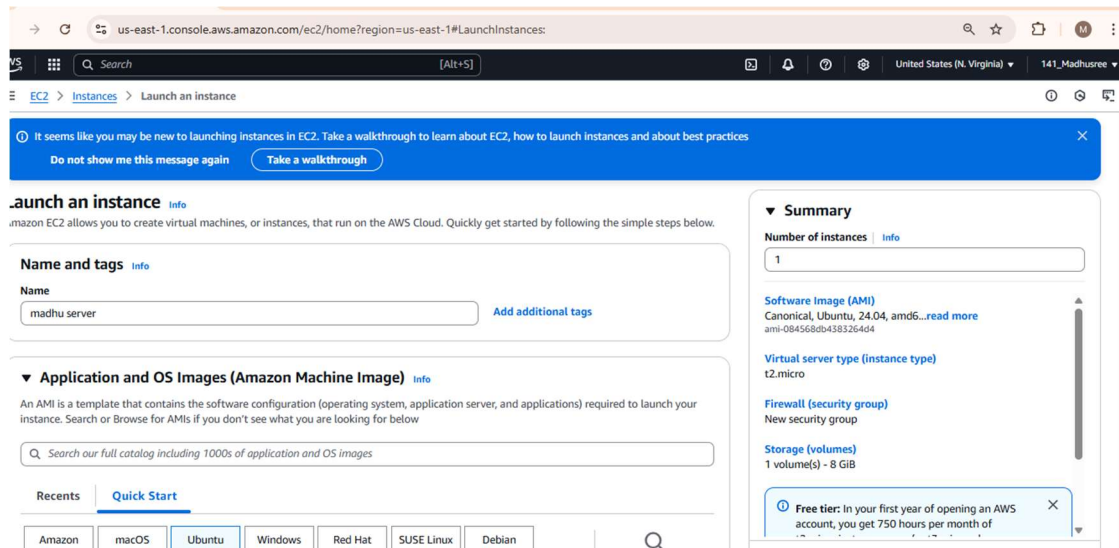
Step 1:

Upload required files to GitHub

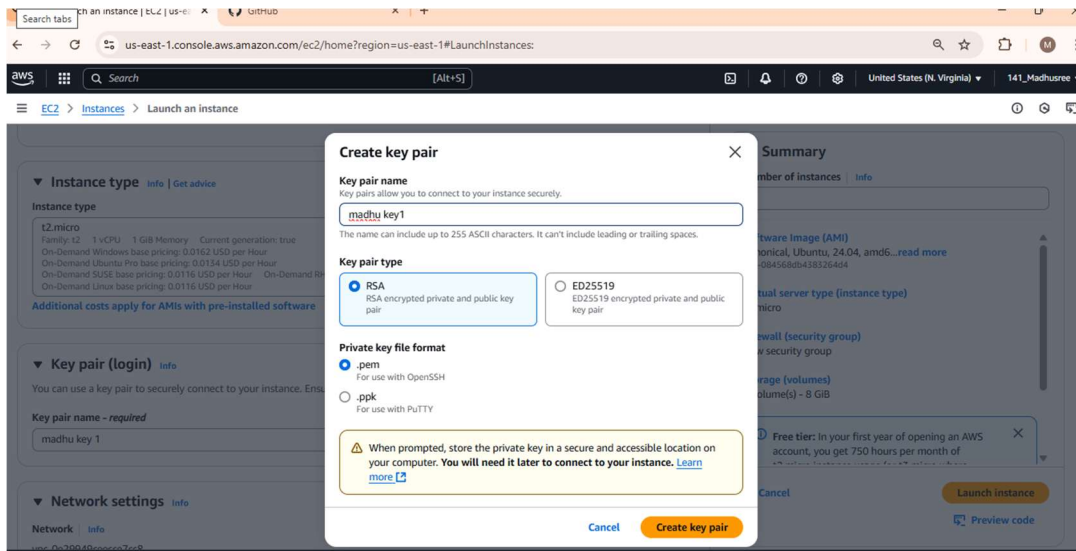


Step 2:

Log into AWS and open EC2. Click on launch instance. Give a name to the instance and select the operating system as Ubuntu.

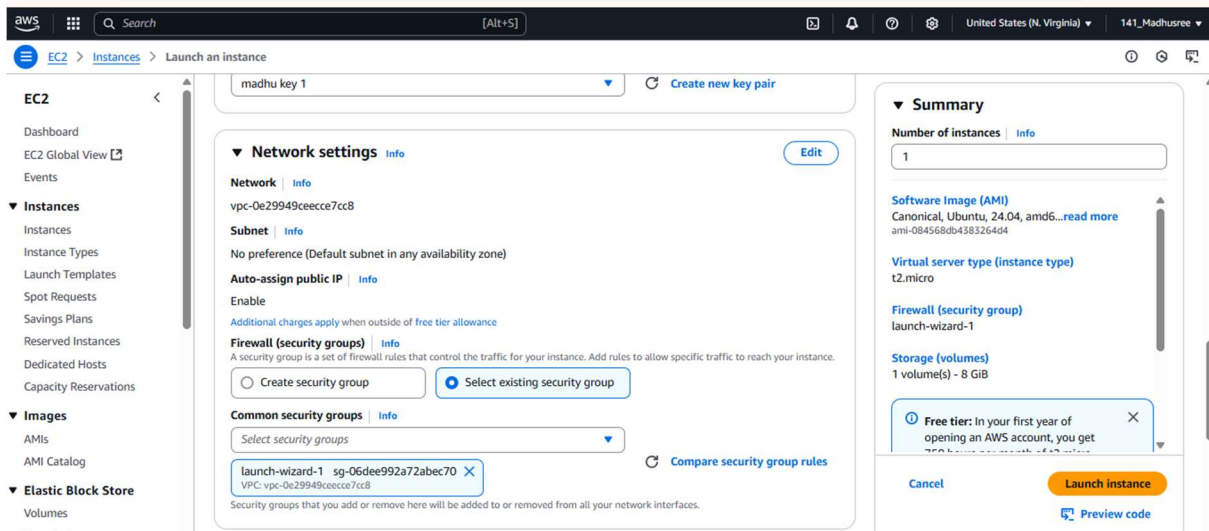


Step 3: Create Key Pair.

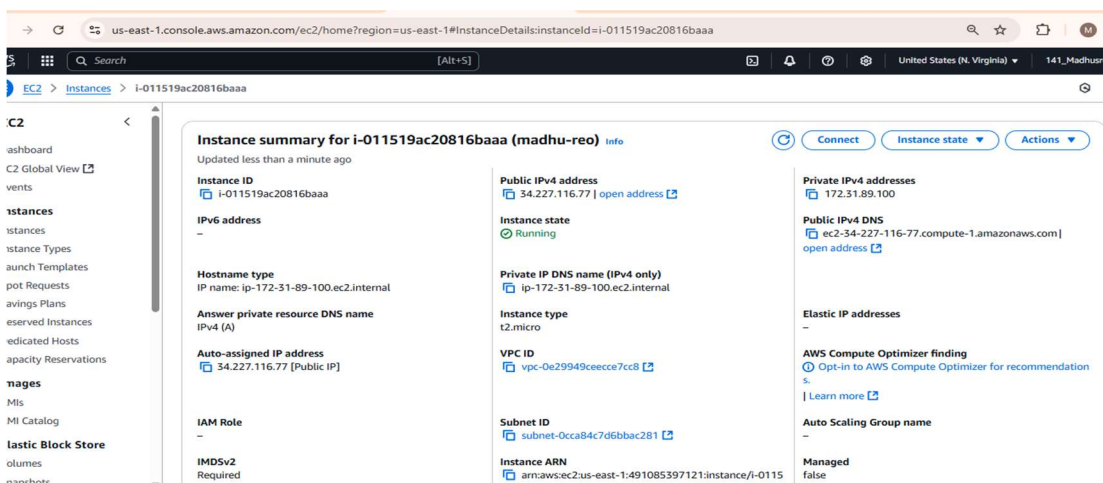


Step 4:

Check the 3 boxes to allow SSH, HTTP and HTTPS traffics and click launch instance.

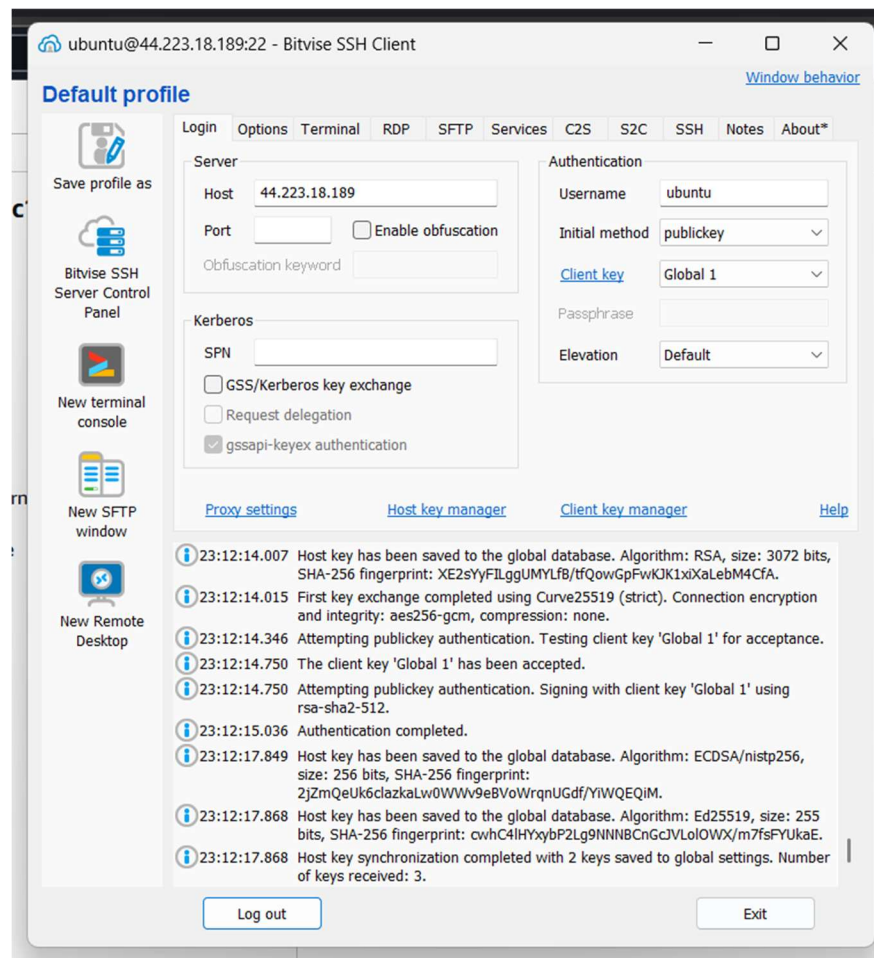


Step 5: Open instances and click of the instance Id of the instance. Copy the IPv4 address.



Step 6:

Open Bitwise SSH Client and paste the IP address in host and set up authentication using the key pair then login and after logging in open new terminal window.



Step 7:

Then Execute command `sudo apt-get update` and `sudo apt-get upgrade` in the Bitwise SSH Client Terminal.

```
ubuntu@ip-172-31-24-135:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-24-135:~$ sudo apt-get update
```

```
Reading package lists... Done
ubuntu@ip-172-31-81-6:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Step 8:

Now Execute `sudo apt -get install nginx` Cmd for setting up the web server.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-81-6:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
```

Step 9:

After that, then open the directory that is cloned using command `cd` name of the directory. Then install NodeJS using commands `curl`:

`https://deb.nodesource.com/setup_18.x | sudo -E bash -` and `sudo apt-get install nodejs`.

```
ubuntu@ip-172-31-24-135:~$ curl -SL https://deb.nodesource.com/setup_18.x | sudo -E bash -
Reading package lists... Done
2025-04-23 15:06:03 - Repository configured successfully.
2025-04-23 15:06:03 - To install Node.js, run: apt-get install nodejs -y
2025-04-23 15:06:03 - You can use N|solid Runtime as a node.js alternative
2025-04-23 15:06:03 - To install N|solid Runtime, run: apt-get install nsolid -y
ubuntu@ip-172-31-81-6:~$ sudo apt-get install nodejs
```

```
ubuntu@ip-172-31-81-6:~$ sudo apt-get install nodejs
Reading package lists... Done
```

Step 10: Now clone the GitHub repository from GitHub using command `git clone` paste the link of the repository

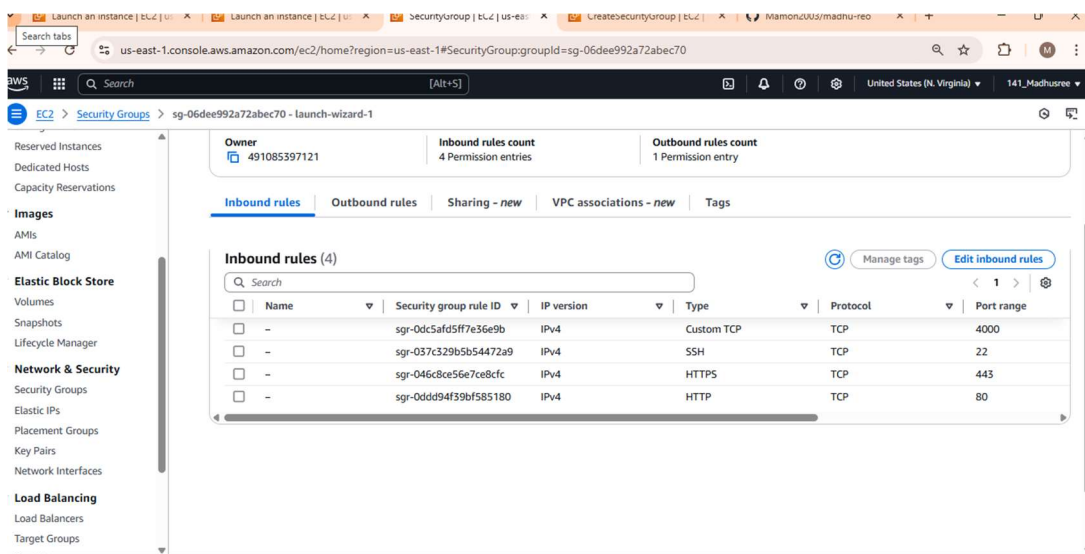
```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-81-6:~$ git clone https://github.com/Mamon2003/madhu-reo.git
Cloning into 'madhu-reo'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
```

```
ubuntu@ip-172-31-81-6:~$ cd madhu-reo
ubuntu@ip-172-31-81-6:~/madhu-reo$ ls
index.js  package.json
ubuntu@ip-172-31-81-6:~/madhu-reo$ npm install
```

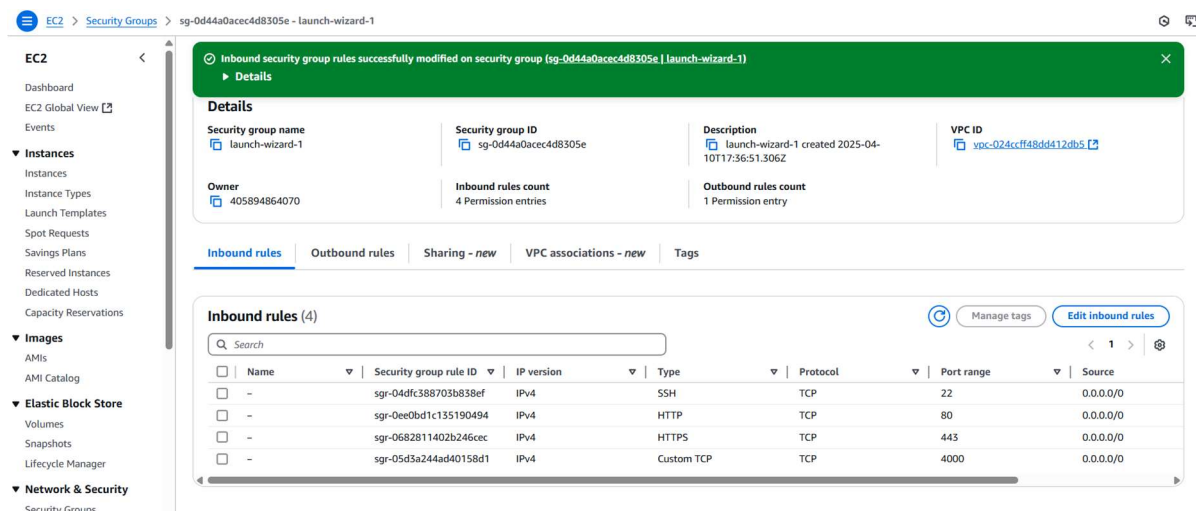
Step 11: After installing nodejs use `npm install` command to install required dependencies.


```
buntu@ip-172-31-81-6:~/madhu-reo$ npm install
added 227 packages, and audited 228 packages in 11s
5 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.3.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.3.0
npm notice To update run: npm install -g npm@11.3.0
npm notice
buntu@ip-172-31-81-6:~/madhu-reo$ node index.js
started server
```

Step 12: Now open AWS and open the current instance then scroll down and go to security Section. Then click on the link in security groups then click on edit inbound rules.



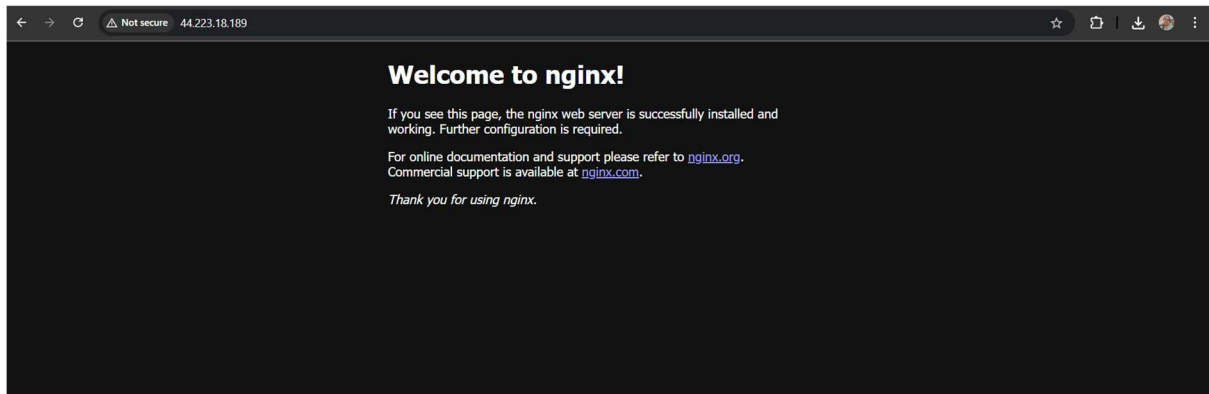
Step 13: Click on add rule and select the type of new rule custom TCP Port range 4000 source 0.0.0.0 or anywhere then click on save rule.



Step 14: Open the Bitvise SSH Client terminal console and start the server using command `node index.js`

```
ubuntu@ip-172-31-81-6:~/madhu-reo$ node index.js
Started server
```

Step 15: Now copy the IPv4 address of the instance and paste it on web browser.



Step 16:

Then, put the port like this IP address: Port (3.110.158.144:3000)

